

Course Code	Course Title	L	T	P	C
BCSE406L	NoSQL Databases	3	0	0	3
Pre-requisite	NIL	Syllabus version			
		1.0			
Course Objectives					
1. To recognize the emergence, requirements and benefits of a NoSQL database. 2. To compare NoSQL databases with each other and relational systems. 3. To create wide-column, document, key-value, graph and object-oriented databases, add content, and run queries.					
Course Outcomes					
1. Define, compare and use the four types of NoSQL Databases. 2. Explain key value databases and apply queries on those databases. 3. Explain the detailed architecture, define objects, load data, query data and performance tune Document-oriented NoSQL databases. 4. Demonstrate an understanding of the detailed architecture, define objects, load data, query data and performance tune Column-oriented and Graph oriented NoSQL databases. 5. Evaluate NoSQL database development tools and programming languages.					
Module:1	Introduction	5 hours			
The Relational Database Revolution with SQL - Design Limitations - Impedance Mismatch - Schema Evolution - Horizontal Scalability- Motivations for Not Just/No SQL (NoSQL) Databases - Data Management with Distributed Databases – SQL Versus NoSQL Databases - The CAP Theorem - ACID and BASE - Types of Eventual Consistency - Types of NoSQL Databases					
Module:2	Key-Value Databases	4 hours			
From Arrays to Key-Value Databases - Essential Features of Key-Value Databases - Properties of keys - Characteristics of values - Data Modeling Terms for Key- Value Databases - Key - Architecture and Implementation terms - Designing Structured Values - Limitations of Key-Value Databases - Design Patterns for Key-Value Databases - Redis database - Queries - Case Study: Key-Value Databases for Mobile Application Configuration.					
Module:3	Document Databases	6 hours			
Introduction to Document Databases – Mongo DB - Collections - Basic operations on collections (CRUD) - Find operation - Sorting - Limiting - Aggregate operations: Aggregation pipeline - Operators - Combining aggregate operators					
Module:4	Designing Document Databases	7 hours			
Partitioning - Types of Partitions - Vertical Partitioning - Horizontal Partitioning or Sharding - Separating Data with Shard Keys – Replication - Distributing Data with a Partitioning Algorithm - Data Modelling and Query Processing - Normalization – De-Normalization - Query Processor - Indexing - Distributed Consistency - Joins					
Module:5	Column Family Databases	6 hours			
Introduction to Column Family Databases - Google Big Table - Differences and Similarities to Key-Value and Document Databases - Architectures Used in Column Family Databases - Cassandra Architecture: Peer-to-Peer - Commit Log - Bloom Filter - Consistency Level - Processes and Protocols - Replication - Anti-Entropy					

Gossip Protocol - Hinted Handoff - Handling of deletion - When to Use Column Family			
Module:6	Designing Column Family Databases		7 hours
Column Family Database Terminology – Key space - Row Key - Column - Column Families – Cassandra - CQL Queries - Primary Key and Clustering Key - CRUD operations - Cluster - Partition - Replication - Consistency levels - Guidelines for Designing Tables - Indexing - Primary and Secondary Index			
Module:7	Graph Databases		8 hours
Introduction to Graph Databases - What Is a Graph? - Graphs and Network Modeling - Advantages of Graph Database - Elements of Graphs - Operations on Graphs - Properties of Graphs and Nodes - Types of Graphs - Graph Design - Neo4J - Cypher Query Language: Creating, Removing and Querying Nodes and Relations - Basic Graph Traversal - Finding Path and Distance between nodes - Gremlin: Query by Graph Traversal - Using NoSQL and Relational Databases Together			
Module:8	Recent Trends		2 hours
Guest lectures from Industry and, Research and Development Organizations			
	Total Lecture hours:		45 hours
Text Book(s)			
1.	Dan Sullivan, NoSQL for Mere Mortals, Addison-Wesley Professional, 2015		
Reference Books			
1.	Adam Fowler, NoSQL For Dummies, For Dummies, 1st edition, 2015		
2.	Gerardus Blokdyk, NoSQL Databases A Complete Guide, 5STARCooks, 2021		
3.	Pramod J. Sadalage and Martin Fowler, NoSQL Distilled, Addison-Wesley, 1st edition, 2012		
4.	Guy Harrison, "Next Generation database: NoSQL New SQL and Big Data", Apress, 1 st Edition, 2015.		
Mode of Evaluation: CAT / written assignment / Quiz / FAT			
Recommended by Board of Studies		12-05-2023	
Approved by Academic Council		No. 70	Date 24-06-2023